



Solutions:

Q1: (a)

$$v_e = (1)(8) = 8 \text{ V}$$

(b)

$$i_b = (50)(0.2) = 10 \text{ A}$$

(c)

$$v_c = -(1)(20) = -20 \text{ V}$$

(d)

$$i_d = -50/25 = -2 \text{ A}$$

$$p_{8\Omega} = (8)(28) = (1)(2)(8) = 8 \text{ W}$$

$$p_{50\Omega} = (50)^2(0.2) = 500 \text{ W}$$

$$p_{20\Omega} = (1)^2(20) = 20 \text{ W}$$

$$p_{25\Omega} = (-2)^2(25) = 100 \text{ W}$$

Q2:

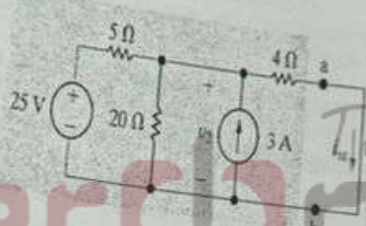
Solving for the currents yields
 $-10 + 6i_e = 0$ and $-3i_e + 2i_o + 3i_o = 0$
 $i_e = 1.67 \text{ A}$ and $i_o = 1 \text{ A}$
 Applying Ohm's law to the 3Ω resistor gives the desired voltage:
 $v_o = 3i_o = 3 \text{ V}$

Q3:

Using nodal voltage method:

$$(v_1 - 25)/5 + v_1/20 - 3 = 0$$

$$(v_1 = 32 \text{ V} = V_{Th})$$



$$(v_2 - 25)/5 + v_2/20 - 3 + v_2/4 = 0$$

$$v_2 = 16 \text{ V}$$

Hence, the short-circuit current is
 $I_{sc} = 16/4 = 4 \text{ A}$

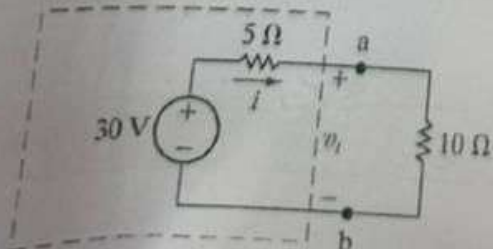
$$R_{Th} = V_{Th} / I_{sc} = 32/4 = 8 \Omega$$

Q4:

From A to C: R_3 is in parallel with the series combination of R_1 and R_2
 $R_3 \parallel (R_1 + R_2)$

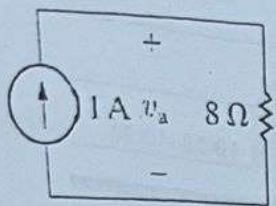
From B to C: R_2 is in parallel with the series combination of R_1 and R_3 .
 $R_2 \parallel (R_1 + R_3)$

Q5:

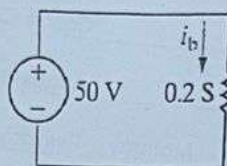


a)

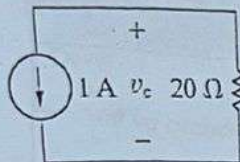
$$b) p_{10\Omega} = (2)(2)(10) = 40 \text{ W}$$



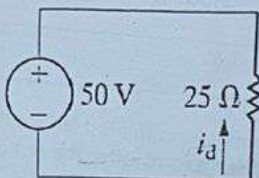
(a)



(b)



(c)



(d)

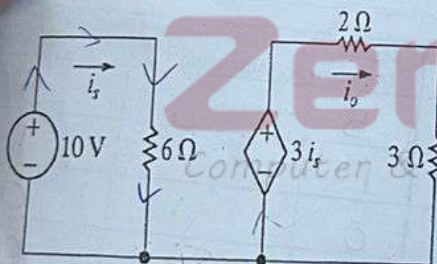
Q1: Calculate the following values for the above circuits: (8 marks)

- (a) $v_a = \dots 8V$, $p_{8\Omega} = \dots 8W$
 (b) $i_b = \dots 1A$, $p_{0.2S} = \dots 50W$
 (c) $v_c = \dots 20V$, $p_{20\Omega} = \dots 20W$
 (d) $i_d = \dots 2A$, $p_{25\Omega} = \dots 100W$

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Use KVL & Ohm's law to find the voltage v_o

(4 marks)



Solution:

$$V = IR$$

$$10 = i_s \times 6$$

$$i_s = \frac{10}{6} A$$

$$3i_s = 5A$$

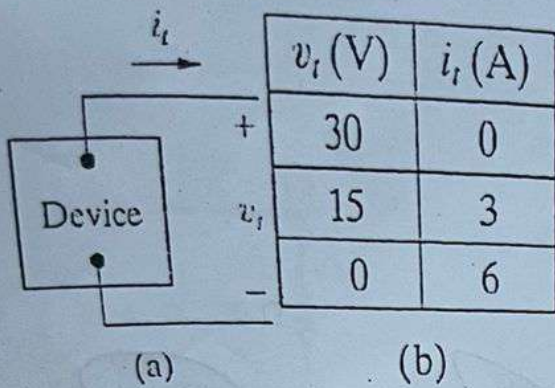
$$3i_s = i_o = 5A$$

$$KVL: -5 - 5(2) + 3v_o = 0$$

$$v_o = \frac{15}{3} = 5V$$

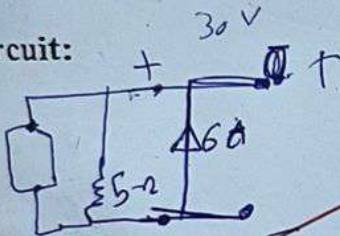
Q5:

(4 marks)



1. Construct a circuit model of the device inside the box.
2. Using this circuit model, predict the power this device will deliver to a $10\ \Omega$ resistor.

1. The circuit:



2. $p = i^2 R = 36 \times 10 = 360\text{ W}$

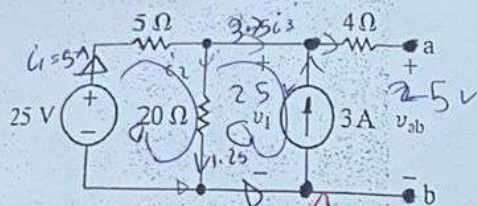
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Q3: Find the Thevenin equivalent voltage of the circuit in Fig and calculate R_{TH} using: $V_{TH}/I_{s,c}$

(10 marks)



1. Calculating V_{TH} :

$$V_T = 25V \text{ parallel}$$

2. Finding R_{TH} , using $R_{TH} = V_{TH} / I_{s,c}$

$$R_{TH} = \frac{V_{TH}}{I_{s,c}} = \frac{25}{3.75} = 6.66 \Omega$$

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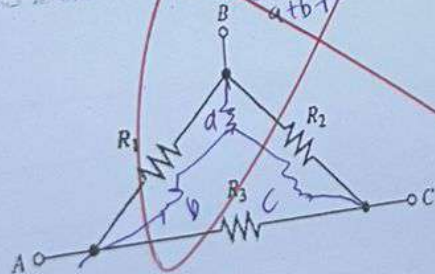
Q4: Describe the total resistance between each pair of terminals in Figure

1. From A to C:

$$R_3 + \frac{a}{a+b+c} \Omega$$

2. From B to C:

$$\frac{a+b}{a+b+c} \Omega$$



(4 marks)